
GIT PUSH THE BEATS

Melody Dive Use-Case Specification: Sign In

Version 1.0

Melody Dive	Version: 1.0
Use-Case Specification: Sign In	Date: 15/11/2025
UC SignIn	

Revision History

Date	Version	Description	Author
15/11/2025	1.0	Add: Add detail information about this use case.	Thái Minh Huy

Melody Dive	Version: 1.0
Use-Case Specification: Sign In	Date: 15/11/2025
UC SignIn	

Table of Contents

1. Use-Case Name: Sign In	4
1.1 Brief Description	4
2. Flow of Events	4
2.1 Basic Flow	4
2.2 Alternative Flows	5
2.2.1 Invalid Credentials	5
2.2.2 Forgot Password	5
3. Special Requirements	5
3.1 Login verification must complete within 3 seconds.	5
3.2 The system must securely hash and compare passwords.	5
3.3 The login interface must be responsive and mobile-friendly:	5
4. Preconditions	6
4.1 The user is currently not logged into the system.	6
4.2 The user has already registered an account using the “Sign Up” use case.	6
5. Postconditions	6
5.1 The user is successfully authenticated.	6
5.2 The system redirects the user to their appropriate dashboard (User, Premium User or Admin).	6
5.3 A session or authentication token is created for the user.	6
6. Extension Points	6
6.1 Sign Up	6

Melody Dive	Version: 1.0
Use-Case Specification: Sign In	Date: 15/11/2025
UC SignIn	

Use-Case Specification: Sign In

1. Use-Case Name: Sign In

1.1 Brief Description

The “Login” use case allows an unlogged user to access the system by providing valid login credentials (username and password). If the user doesn’t have an account, they may extend this use case by navigating to the “Sign Up” use case.

2. Flow of Events

2.1 Basic Flow

1. The user navigates to the login interface.
2. The system displays a form requesting username and password.

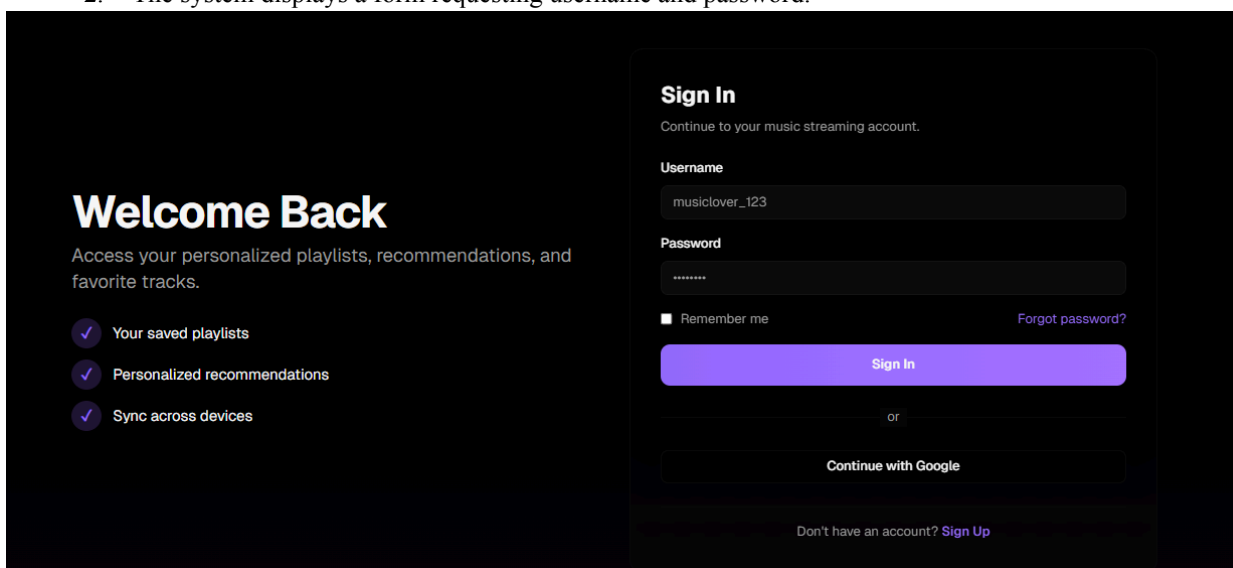


Figure 1: The interface of Sign In

3. The user enters their username and password.
4. The system verifies the credentials in the database:
 - If valid, the system logs the user in and redirects them to their appropriate homepage(User, Premium User or Admin).
 - If invalid, the system displays an error message and asks the user to try again.

Melody Dive	Version: 1.0
Use-Case Specification: Sign In	Date: 15/11/2025
UC SignIn	

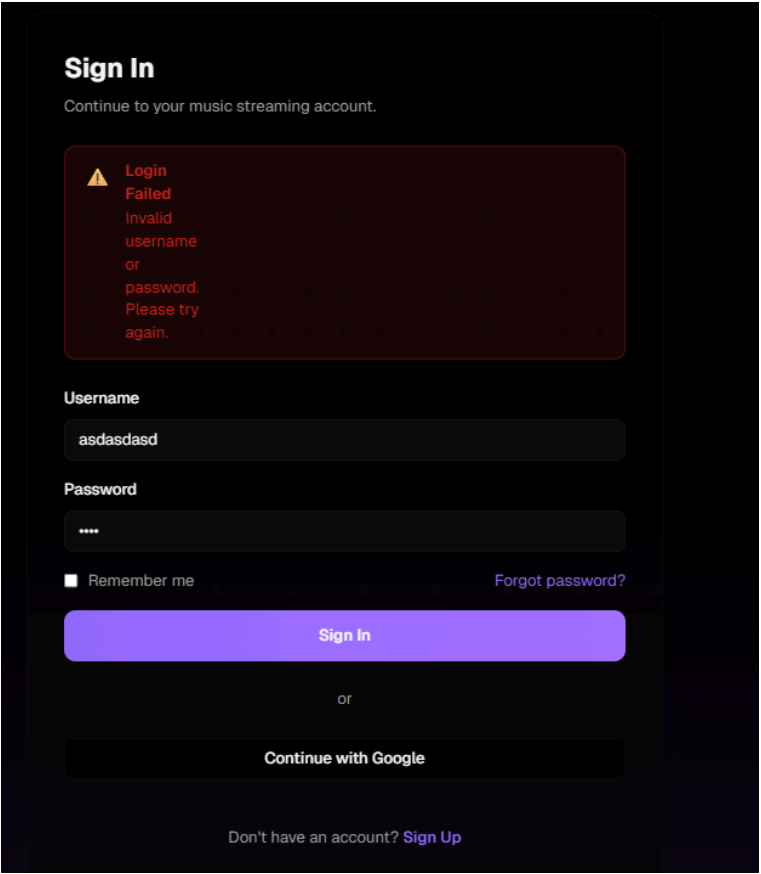


Figure 2: The interface of Login when failed

5. When successful, the user gains access to the system’s functionalities.

2.2 Alternative Flows

2.2.1 Invalid Credentials

- If the user provides an incorrect username or password.
- The system displays an error: “Invalid username or password”.
 - The flow returns to step [2.1.2] for retry.

2.2.2 Forgot Password

- If the user clicks “Forgot Password”:
- The system redirects to the password reset process.
 - “Sign In” use case is paused and resumes after a successful password reset.

3. Special Requirements

3.1 Login verification time

The login verification process must complete within 3 seconds.

4. Preconditions

4.1 The user is currently not logged into the system.

The user is not logged in an account yet, and still is at the login interface.

Melody Dive	Version: 1.0
Use-Case Specification: Sign In	Date: 15/11/2025
UC SignIn	

5. Postconditions

- 5.1 The user is successfully authenticated.
- 5.2 The system redirects the user to their appropriate dashboard (User, Premium User or Admin).
- 5.3 A session or authentication token is created for the user.

6. Extension Points

6.1 Sign Up

If the user does not have an account, they may extend this use case by clicking “Sign Up” and completing the registration process.